Table X.X.1 **EPTControl** Cluster of control parameters for configuring engine position tracking for a specific pattern type.

Type	**Variable Name**	**Description**	**Value during lab operation**
U16	NumberOfCrankTeeth	Specifies the number of teeth on the crankshaft trigger wheel as if there were no missing or extra crank teeth. For example, a 60-2 crankshaft pattern would have NumberOfCrankTeeth = 60.	3,600
TF	Stroke (FALSE=2, TRUE=4)	When FALSE, the EPT is tracking teeth from 0 to NumberOfCrankTeeth. When TRUE, the EPT is tracking teeth from 0 to 2*NumberOfCrankTeeth.	TRUE
U32	CrankStallPeriod	The maximum crank tooth period between evenly spaced crank teeth in terms of 40 MHz clock ticks. A measured crank tooth period greater than CrankStallPeriod causes the Period output of the EPT VI to report as zero and the CrankStalled output to report as TRUE. Engine speed in RPM can be converted to CrankStallPeriod in 40 MHz clock ticks according to the following equation: CrankStallPeriod(clock ticks) = (60 * 4E7) / (EngineSpeed(RPM) * NumberOfCrankTeeth)	5,000,000 (0.1333 RPM)
U16	CrankCountStart	Specifies the number of teeth on the crankshaft trigger wheel to count before looking for the pattern-specific unique feature indicating tooth 0. This can help with waiting for a cam shaft VR signal to gain recognition, or it can help reject a number of false triggers during the initial engagement of a starter motor. The value must be set with a minimum of 2.	3,600

U16	PeriodAccumCount	Specifies the number of crank teeth to accumulate Period in the PeriodAccum output of the EPT VI. PeriodAccum can be used to measure an average crankshaft speed over multiple crank teeth.	3,600
TF	SyncEnable	When TRUE, sync is enabled and position tracking will take place if a valid crank and cam signal pattern is presented to the EPT VI at an engine speed greater than the stall speed.	TRUE
TF	SimEnable	When TRUE, CrankSig and CamSig inputs within the EPTInSig cluster are internally disconnected from the position tracking core and simulated signals are connected in their place.	FALSE
TF	MissedCrankFlagClr	When TRUE, the MissedCrankFlag within the EPTData cluster will be cleared.	
TF	MissedCamFlagClr	When TRUE, the MissedCamFlag within the EPTData cluster will be cleared.	
U16	SimPeriod	The simulated period between evenly spaced crank teeth in terms of 40 MHz clock ticks. SimPeriod is only effective when it is set to a non-zero value while SimEnable is TRUE. The EPT will not sync to the simulator unless the effective engine speed is greater than the stall speed. Engine speed in RPM can be converted to SimPeriod in 40 MHz clock ticks according to the following equation: SimPeriod(clock ticks) = (60 * 4E7) / (EngineSpeed(RPM) * NumberOfCrankTeeth)	N/A
TF	WatchdogIn	The WatchdogIn Boolean signal must be toggled at a rate greater than 4 Hz. The purpose of the watchdog is to shutdown engine position tracking (as well as any engine synchronous outputs) in the case of a loss of LabVIEW RT communication with the FPGA. For example, the LabVIEW RT application may	This is toggled at a 1kHz rate by APC_9059_

		fail due to coding errors without properly closing the FPGA application. It would not be desirable for the EPT to continue tracking engine position and supervising engine synchronous outputs according to the last command received from the LabVIEW RT application. The WatchdogIn can be toggled by a simple NOT function and feedback node within LabVIEW RT.	TS10ms_loop.vi

Table X.X.2. **EPTInSig** Cluster of crank and cam input signals used by the EPT for tracking engine position.

Type	**Variable Name**	**Description**	**Inputs**
TF	CrankSig	The digital input signal representing the crankshaft trigger wheel pulse train. This input should be wired from a digital input channel of a NI 9411 Differential Digital Input module or a NI 9752 Automotive AD Combo Module. Additional filtering or modification may be inserted before the signal is wired to the EPT VI. The EPT VI responds to the rising edge of CrankSig.	A signal: Mod3/DIO0 B signal: Mod3/DIO1 Z signal: Mod3/DIO2 (cRIO 9049; NI 9401)
TF	CamSig	The digital input signal representing the camshaft trigger wheel pulse train. This input should be wired from a digital input channel of a NI 9411 Differential Digital Input module or a NI 9752 Automotive AD Combo Module. Additional filtering or modification may be inserted before the signal is wired to the EPT VI. The EPT Encoder VI responds to the rising edge of CamSig.	Mod3/DIO3 (cRIO 9049; NI 9401)

Table X.X.3. **EPTData** Cluster of status information related to engine position tracking.

Type	**Variable Name**	**Description**	**Expected Value**
TF	CrankSigOS	A 40 MHz one-clock one-shot generated at the rising edge of each received or simulated CrankSig input. This signal can be used in conjunction with	-

		CrankCount for any purpose within LabVIEW FPGA logic to perform engine synchronous tasks.	
TF	CrankStalled	Set to TRUE when the engine speed is below stall speed.	FALSE
TF	SyncStopped	Set to TRUE while position tracking is not taking place. In other words, no sync.	FALSE
TF	MissedCrankFlag	Set to TRUE upon the detection of a CamSig rising edge while the CrankCount value has not yet reached the expected value. In other words, the **Cam tooth has been detected sooner than expected**. This condition causes a loss of sync. Re-sync is not allowed until the flag is cleared by setting MissedCrankFlagClr to TRUE.	FALSE
TF	MissedCamFlag	Set to TRUE upon CrankCount being incremented past the expected value for the location of a CamSig rising edge before the CamSig rising edge is detected. In other words, the **cam tooth was not received when expected**. This condition causes a loss of sync. Re-sync is not allowed until the flag is cleared by setting MissedCamFlagClr to TRUE.	FALSE
U16	Period	The period between the last two crank trigger teeth, as if there were no missing or extra teeth, in terms of 40 MHz clock ticks. Period will be set to 0 while Stalled is TRUE. Period in clock ticks can be converted to instantaneous engine speed in RPM according to the following equation: EngineSpeedInstant(RPM) = (60 * 4E7) / (Period * NumberOfCrankTeeth)	-
U16	PeriodAccum	The accumulated period over PeriodAccumCount crank teeth in terms of 40 MHz clock ticks. PeriodAccum will be set to 0 while Stalled is TRUE.	-

		PeriodAccum in clock ticks can be converted to average engine speed in RPM according to the following equation: EngineSpeedAve(RPM) = (60 * 4E7) / (PeriodAccum * NumberOfCrankTeeth / PeriodAccumCount)	
U16	CrankCount	The latest tooth count referenced to crank tooth 0. Tooth 0 is determined according to the pattern that is being tracked and is discussed within the pattern specific documentation. CrankCount is provided in terms of a complete engine cycle and is reported as if there were no missing or extra teeth in the pattern. For example, if PATTERN = N-M, NumberOfCrankTeeth = 60 and Stroke = TRUE, then CrankCount will range from 0 to 119.	0 to 7199
U16	CurrentPositi on	During sync, the EPT provides the **latest crankshaft position in terms of absolute Crank Angle Ticks (CAT) over a complete engine cycle**, referenced to position 0. Position 0 is determined according to the pattern that is being tracked and is discussed within the Pattern-Specific Information section of the EPT documentation. The CurrentPosition in CAD can be calculated according to the following: CurrentPosition(CAD) = [CurrentPosition(CAT) * Stroke * 360] / MAX_CAT Where Stroke = 2 if 4-stroke and Stroke = 1 if 2-stroke.	0 to 28799
U16	MAX_CAT	**Maximum Crank Angle Ticks per engine cycle**. A Crank Angle Tick (CAT) is a single unit of angular measure reported by the CurrentPosition output of the EPT VI. Reported as a power-of-two angular ticks of crank position travel, having a resolution dependent on EXTRAP and the number of physical teeth per crankshaft rotation. For example, if using the N-M EPT VI, which has an extrapolation of 7, the number of CAT per crank tooth would be 2^7=128, and	28,800

		CurrentPosition would be evenly incremented by 128 CAT from one physical tooth to the next. If a 60-2 pattern were used, the maximum number of CAT per crankshaft rotation (cycle) would be 60*128=7680. If the engine was a 4-stroke, the total number of CAT per engine cycle would be 2*60*128=15360. For the CAT C9 test unit, MAX_CAT = 3,600*2^2*2 = 28,800	

Table X.X.4.  **EPTOutSig**: Cluster of crank and cam outputs. These signals are connected directly to the inputs during normal operation, and connected to the simulated signals while SimEnable is TRUE.

Type	**Variable Name**	**Description**
	CrankSigOut	Digital output signal originating from two different sources. If EPT simulation is disabled, then the crank and cam signal inputs are passed directly through to these outputs. If EPT simulation is enabled, then the simulated signals are output here.

	CamSigOut	Digital output signal originating from two different sources. If EPT simulation is disabled, then the crank and cam signal inputs are passed directly through to these outputs. If EPT simulation is enabled, then the simulated signals are output here.

FuelSparkSupervisor: This cluster output must be wired directly to the FuelSparkSupervisor cluster inputs of NI Powertrain Controls fuel and/or spark VIs.

APC_9049_TS10ms_loop.vi

Table Y.X.1. **PI_Gens_Time_Config_In**: The pulse generation configuration cluster is a group of parameters to configure a (up to) 5-pulse injection sequence. The duration and timing parameters are converted from milliseconds and angles to 40

MHz clock ticks and Crank Angle Ticks, respectively, within this VI to be sent to the FPGA level via the PlsGen_Config_Out cluster to one of the channels of the PlsGen_StdT5 VI.

Type	**Variable Name**	**Description**	**Value**
SGL	Window_Start (DBTDC)	The injection pulse sequence for each channel should be bounded by an angle-based window, defined by a Window_Start angle and a Window_End angle. The angle windows for each channel must not overlap. The units of Window_Start and Window_End at the RT level are Degrees Before TDC (DBTDC).	200
SGL	Window_End (DBTDC)		0
TF	MainEnable	Enables the Main pulse.	Set by PC_ControlSettings '**DI enable**'
SGL	MainDuration (msec)	Determines the length of the Main fuel pulse. MainDuration is entered in milliseconds. The DI Driver FPGA Express VI device driver limits any pulse to 5 milliseconds by default.	Set by PC_ControlSettings '**DI duration [ms]**'
SGL	MainSOI (DBTDC)	The main fuel pulse is generated with a leading edge coinciding with Main_SOI in units of Degrees Before TDC (DBTDC). The length of the pulse will be according to MainDuration.	Set by PC_ControlSettings '**DI advance [CABTDC]**'

ept_enc_vte2_revc VI: Engine Position Tracking VI Pattern Type = Encoder Extrapolation Level = 2 Bits. To be placed within a single cycle loop of an FPGA block diagram, executed at 40 MHz. See the Pattern-Specific Information section in the EPT documentation